

UNIT 7

A HISTORY OF AUTOMATION: THE RISE OF ROBOTS AND AI

Over time, we've grown reliant on automated technology. It's found in almost every part of our lives, from automatic doors to factory line robots, to business process automation.

Nowadays, artificial intelligence is the talk of the town, and the dreaded robot takeover seems to be looming ever closer. We have customer service handling chatbots, AI in our back pockets, and increasingly 'smart' homes.

But how did we get here? We explore the history of automation and the rise of robots and AI.

Early machines and concepts

The concept of AI isn't as new as you'd think. Some suggest that the idea of artificial intelligence was conceived by the Ancient Greeks. However, without tangible machines and computers, we wouldn't have developed any kind of mechanical automation. And so, the history of automation starts far closer than Aristotle and Socrates.

The early life of automation begins, then, with the industrial revolution and industrial machinery between 1790 and 1840. Then, like now, people feared the impact of automation on their jobs (though it all turned out well in the end).

In 1837, Charles Babbage began the creation of a prototype machine he called The Analytical Engine, which was the first device to earn the name 'computer'. His friend Ada Lovelace, meanwhile, created the first-ever computer program, which would have run on the machine. Sadly, Babbage died before his prototype was complete.

Creation of robots

Early on, the concept and development of robots, automation, and AI went unrecognised by most of the world. Still, people of many different backgrounds, cultures, and professions studied the obscure interest.

In the early 1900s, the term robot was coined by Karel Capek, a Czech writer, in his 1920 play *Rossum's Universal Robots*. Having coined the term, Capek's play also presented the first instance of robots taking over the world. (The science fiction trope started alongside the creation of robots themselves.)

By 1939, the first physical robot, ELEKTRO, went on display at the World's Fair — marking a key point in the history of automation. With human commands, he could walk, smoke cigarettes, and blow up balloons.

Three laws of robotics

Following this, in the early 1940s, Isaac Asimov created the three laws of robotics, which he used in his works of fiction. These rules were then accepted by other writers who used them in their science fiction pieces as well. The three rules dictate the ways a robot must act with regard to humans:

One: A robot may not injure a human being or, through inaction, allow a human being to come to harm.

Two: A robot must obey orders given by human beings except where such orders would conflict with the First Law.

Three: A robot must protect its existence as long as such protection does not conflict with the First or Second Law.

First AI

With the concept of robots invented and generally understood by most, the first autonomous robots in the history of automation were created in 1948 by William Grey Walter. These came in the shape of two 'tortoises': Elmer and Elsie. The robots could find their way around obstacles without human help by following the light and making use of a bump sensor. Remarkably, the success in the interaction of these two 'sensory' inputs (light and touch) which enabled these robots to work, also helped us to better understand our nervous system.

In 1950, Alan Turing devised a way to measure the 'intelligence' of a machine, which tested the machine's ability to 'think'. It is known as The Turing Test and is still used today.

Academic recognition

With so much success in the world of robotics, the study started to gain recognition within the academic community. In 1956, a Dartmouth University conference introduced the term 'artificial intelligence' and the field of AI research.

This encouraged many to continue the development of robotics and officially started research into artificial intelligence development.

The history of automation is not fully successful, however. Negative reports about Natural Language Processing (NLP) in 1966, which highlighted the failings of a 10-year study, caused a delay in the further development of language processing in AI. It wasn't until the early 70s that an influential group focused on Natural Language Processing was established at the Artificial Intelligence Centre at the Stanford Research Institute.

The 60s, 70s, and 80s

Notable robots in the 60s included the 1966 therapy chatbot 'ELIZA', who had started life as a joke. Then there was the 1968 mobile robot affectionately named 'Shakey'.

Following this, the history of automation saw something of a subdued period. The 70s and 80s came to be referred to as the AI winter. Interest in the development of AI dwindled, and people became more pessimistic about its chances of success. This didn't mean that AI research was on hold, merely that it stepped out of the limelight for a while.

We did see some developments, though. There was SCARA the assembly line aid in 1979. There was the robot that learnt from experience, RB5X, in 1984. Then, in 1985 the term 'virtual reality' was coined with the first sales of VR gloves and glasses. The phrase 'augmented reality' would shortly follow, invented in 1990 by Tom Caudell. Around that same time, (1989) Sir Tim Berners-Lee invented hyperlinks and hypertext and used them to invent the World Wide Web.

Computerised automation started to develop around this time, and the 90s would mark the start of AI development moving away from physical bots to digital programs.

90s AI and automation software

The 1990s was a time for major advances in AI. An artificial intelligence called 'Deep Blue' defeated chess grandmaster Garry Kasparov at chess. NASA deployed its first autonomous robotics system, Sojourner, on the surface of Mars.

Meanwhile, web crawlers (and other AI-based data extraction programs) became a key part of widespread web use.

The 90s also marks the time when the history of automation fully intertwined with business process management. The growth of BPM and computerised automation led to the 90s being known for large BPM systems.

The millennium

The history of automation suffered a distinct gap in the advancement of technology in the early 2000s. After the creation of ASIMO, (which is claimed to be 'the world's most advanced humanoid robot') by Honda in 2000, automation development went quiet for a while.

Negative reviews of business process management (for which business process automation is an invaluable tool) saw search rates for the term drop by more than half between 2004 and 2011. Business interest in automated management solutions had unexpectedly dipped.

In 2011, the release of Apple's Siri broke the radio silence period regarding automation breakthroughs. Siri triggered a new age of automation and AI-driven assistants. This embodied the move away from physical robots and into the computerised automation development, and AI software that had begun in the late 80s and early 90s.

The present day

Business process automation or BPA (recently becoming known as robotic process automation or RPA) is becoming more and more refined and efficient. In the present day, automation software has become a necessity rather than a luxury. Its widespread use is optimising employee time and work, leading to enormous resource savings.

We experience the wonder of AI daily, be it on Twitter, in emails, in our video games, or elsewhere. We have artificial intelligence assistants in our phones, our cars, and our homes. All this in the youth of the technology, too.

It isn't perfect. Siri doesn't always have the answer, Alexa occasionally mishears us. NPCs in video games do stupid (and hilarious) things, and as recently as 2016 we saw the failure that was Microsoft's Tay.

But despite these shortfalls, AI and automation are more versatile now than ever before. And they're developing and improving by the day.

The future of automation

The history of automation, despite a few bumps along the way, has seen a lot of success in a short period. It continues to grow and evolve today, providing us with more innovative solutions, interactive AI, and assistance in unravelling the secrets of the universe.

It's impossible to know if Karel Capek and all the science fiction writers after him were right about a future robot rebellion. What's clear is that the future looks to be automated. As exciting as current and future tech is, we shouldn't forget the history of automation or the work that it has taken to get us where we are today.